Should Platforms Be Allowed to Sell on Their Own Marketplaces?

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Author: Andrei Hagiu, Tat-How Teh, and Julian Wright

Reporters: Jinghao Mu, Jialong Zheng, Binghui Guo, Zhiyuan Chen, Zhuoyue Wu

Lingnan College, Sun Yat-Sen University

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PART 1

Research Question

Presenter: Jinghao Mu

Advanced Microeconomics Presentation

Motivation: A Dual Role platform

Many large platforms now play two roles at the same time.
(Eg. Amazon, JD.com, Apple App Store...)

- They run a marketplace for third-party sellers.
- They also sell their own products on the same marketplace.

The key tension

The platform is both the market **organizer** and one of the **sellers** in the market.

Why Dual Mode Is Controversial

Dual mode has both potential harms and potential efficiency benefits.

Regulatory concerns

- **Product imitation:** copy successful sellers using marketplace data.
- **Self-preferring:** steer consumers toward the platform's own products.
- **Innovation harm:** weaken third-party sellers' investment incentives.

Potential benefits

- **One-stop shopping:** consumers can compare more options in one place.
- **Efficient supply:** the platform may be a better seller in some categories.
- **Price pressure:** the platform's product can constrain third-party prices.

Policy question

Should regulators ban dual mode itself, or target specific harmful conduct?

Research Question

The paper asks three linked questions.

1. Does **banning** dual mode always **raise consumer surplus and welfare**?
2. If dual mode is banned, does the platform **switch to seller mode or marketplace mode**?
3. Are targeted **rules on imitation and self-preferring** better than a full ban?

Key Insight

The problem is not dual mode itself, but how the platform uses its power.

- A ban on dual mode is not always good for consumers or welfare.
- Rules that target imitation and self-preferring can work better.

Location and Contribution

The paper is located at the intersection of platform economics and antitrust policy.

- It studies a platform that can choose among **marketplace mode**, **seller mode**, and **dual mode**.
- It connects the policy debate on platform bans with a formal welfare analysis.
- It separates **dual mode itself** from two possible bad conducts: **product imitation** and **self-preferring**.

Presentation Roadmap

The presentation is organized into five parts.

1. Part 1: Motivation and main question — Jinghao Mu
2. Part 2: Baseline model setup — Jialong Zheng
3. Part 3: Baseline equilibria and price squeeze — Binghui Guo
4. Part 4: Ban on dual mode — Zhiyuan Chen
5. Part 5: Imitation, self-preferring, and policy comparison — Zhuoyue Wu

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PART 2

Baseline Model

Presenter: Jialong Zheng

Players I: Consumers and Competitive Benchmark

Consumers

- Outside option of not buying anything (e.g., postpone purchase or buy another product): $v_o \sim G$.[a]
- Platform transaction benefit: $b > 0$ (finding products, one-click payment, delivery tracking, easy returns).
- Choice rule: buy the option with the highest net utility.

[a] G is log-concave and continuously differentiable on $[0, \bar{v}_o]$, with density g.

Fringe sellers

- At least two identical sellers.
- Sell a basic product with value v.
- Consumers already know them and can buy directly.
- Bertrand competition gives the price benchmark.[a]

[a] All marginal costs are normalized to zero.

Players II: Platform and Innovative Seller

Innovative (superior) seller S

- Benefits from innovation; discovered only if listed on M.
- Better product value: $v + \Delta > v > 0$.
- Innovation: chooses quality investment Δ.

Platform M

- Chooses marketplace, seller, or dual mode.
- Seller: sells its own offering, valued at $v + \sigma$.
- Marketplace: sets commission $\tau \geq 0$ for third-party sellers.

Values and Parameters

Product values are:

Product	Consumer value	Meaning
Fringe sellers	v	Basic product
Innovative seller S	$v + \Delta$	Better product
Platform M	$v + \sigma$	Platform's own product

- $b > 0$: convenience benefit from buying through the platform.
- $v_o \sim G$: outside option.
- τ: platform commission.

Intuition

$\sigma > 0$ means the platform is a better seller than fringe sellers; $\sigma < 0$ means the opposite.

Innovation by Seller S

Seller S chooses innovation $\Delta \geq \Delta^l$ and pays fixed cost $K(\Delta)$.

$$K'(\bar{\Delta}) = G(v + \sigma + b)$$

Meaning of $\bar{\Delta}$

$\bar{\Delta}$ is the high innovation level that can arise when the market gives strong rewards to innovation.

Intuition

The left side is marginal cost of innovation. The right side is the demand gain from better quality.

Product Discovery

Consumers do not know about S at the start.

- Consumers know the fringe sellers.
- Consumers may know the platform's own product.
- Consumers discover S only if S is available on the platform.

Why this matters

The platform is not only a sales channel. It is also a discovery channel.

Intuition

If a policy changes the platform's mode, it may also change whether consumers find S.

Three Platform Modes

Mode	What M does	Key feature
Marketplace	Hosts third-party sellers	Earns commission τ
Seller	Sells its own product	No marketplace for S
Dual	Hosts sellers and sells own product	Both roles at once

Intuition

The policy debate is about whether the third mode should be allowed.

Timing

The game is solved by backward induction.

1. M chooses its mode and sets commission τ if there is a marketplace.
2. S decides whether to join the platform and chooses Δ.
3. Sellers set prices.
4. Consumers choose what to buy.

Equilibrium concept

Subgame Perfect Nash Equilibrium.

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PART 3

Equilibrium Analysis

Presenter: Binghui Guo

Marketplace Mode

In marketplace mode, M only charges commission τ.

$$p_i \leq \tau + \Delta, \qquad p_o \leq \tau + \Delta - b$$

Showrooming constraint

If $\tau > b$, seller S can use the platform for discovery and then sell directly.

Intuition

The platform can charge at most the extra convenience value b that it gives consumers.

Marketplace Equilibrium

Proposition 1:

$$\tau^{mkt} = b, \qquad G(\nu) = K'(\Delta^{mkt}), \qquad p_i^* = b + \Delta^{mkt}$$

$$\Pi^{mkt} = bG(\nu), \qquad \pi^{mkt} = \Delta^{mkt} G(\nu) - K(\Delta^{mkt})$$

Intuition

M sets the highest fee that does not push trade outside the platform. S invests until marginal cost equals marginal market gain.

Seller Mode

In seller mode, the platform only sells its own product.

$$\max_{p_m \leq b+\sigma} p_m G(v + b + \sigma - p_m) \tag{3.1}$$

The competitive constraint binds:

$$p_m^* = b + \sigma$$

Proposition 2

M sells to all consumers, while S sells to no one.

Intuition

The platform captures its convenience value and its product advantage, but consumers do not discover S.

Dual Mode: Two Pricing Cases

In dual mode, M both charges commission and sells its own product.

Case 1: $\sigma \geq \Delta$

The platform's own product is good enough. Consumers buy from M.

$$p_i^* = \tau, \qquad p_m^* = \tau + \sigma - \Delta$$

Case 2: $\Delta > \sigma$

Seller S has the better product. But M still puts pressure on S's price.

Intuition

Dual mode matters most when S is better, but M can still discipline S's price.

Price Squeeze

In the price squeeze equilibrium:

$$p_i^* = p_m^* + \Delta - \sigma \tag{3.2}$$

$$p_m^* \in [\max\{\tau - \Delta + \sigma, 0\}, \tau + \min\{\sigma, 0\}]$$

$$\Pi = \tau G(v + \sigma + b - p_m^*)$$

Intuition

M may not sell, but its own product caps S's price. A lower S price raises demand and commission income.

Innovation Constraint

Define $\bar{\tau}$ by:

$$(\bar{\Delta} - \sigma - \bar{\tau})G(\nu + b + \sigma) - K(\bar{\Delta}) = 0$$

Lemma 1:

- If $\tau \leq \bar{\tau}$, then S chooses high innovation: $\Delta = \bar{\Delta}$.
- If $\tau > \bar{\tau}$, then S chooses low innovation: $\Delta = \Delta^l$.

Intuition

A high commission lowers S's margin. If the margin is too low, innovation is not worth the cost.

Dual Mode Equilibrium

Proposition 3: M balances two limits.

- Showrooming constraint: $\tau \le b$.
- Innovation constraint: $\tau \le \bar{\tau}$.

If high innovation is profitable enough:

$$\tau^{dual} = \min\{b, \bar{\tau}\}, \qquad \Delta^{dual} = \bar{\Delta}$$

Otherwise:

$$\tau^{dual} = b, \qquad \Delta^{dual} = \Delta^{l}$$

Intuition

The platform may charge less to keep S willing to innovate, because innovation expands the market.

Which Mode Does the Platform Choose?

Corollary 1:

- M prefers marketplace mode over seller mode iff $\sigma \leq 0$.
- Dual mode is always better for M than marketplace mode.
- There is a threshold $\underline{\sigma} > 0$ such that:

$$\Pi^{dual} > \Pi^{sell} \quad \text{iff} \quad \sigma < \underline{\sigma}$$

Intuition

Dual mode creates more transactions through price squeeze. But if M's own product is much better, M may prefer to sell alone.

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PART 4

Imitation and Self-Preferencing

Presenter: Zhiyuan Chen

Advanced Microeconomics Presentation

What Is Added?

This part adds two practices to the baseline dual-mode model: **product imitation** and **Self-preferencing**.

Product imitation

- **Assumption:** copied product gives surplus $v + \Delta$.
- **Choice:** S: participate or not; then M: imitate or not if S participates.
- **Motivation:** marketplace data helps M identify and copy successful innovations.

Self-preferencing

- **Assumption:** consumers discover S through M's recommendation.
- **Choice:** M: recommend S or not after prices are set.
- **Motivation:** consumers often rely on platform search, ranking, or recommendations to find new products.

Timing: Where Section 4 Changes the Game

Stage 1
M: mode, τ

Stage 2
S: join, Δ
M: imitate?

Stage 3
prices
p_m, p_i, p_o

Stage 4
M: show S?
consumers buy

New: imitation
only if S participates
and M is in dual mode

New: steering
discovery depends on
M's recommendation

Changes from the baseline

- M copying choice: M imitates or not.
- M disclosure choice: M shows S or not.

Marketplace and Seller Mode

Marketplace mode: same commission τ from any marketplace sale.

If M shows S

Buy S through the marketplace iff:

$$\Delta + b - p_i \geq \max\{\Delta - p_o,\ b - \tau,\ 0\} \qquad (4.1)$$

- Left: surplus from S on M.
- Right: other best outside option.

If M hides S

Buy a fringe seller on the marketplace iff:

$$b - \tau \geq 0 \qquad (4.2)$$

- Left: surplus from a fringe seller on M.
- Right: no purchase.

Recommendation rule

- Show S if Eq. (4.1), holds: M earns commission τ.
- Hide S if Eq. (4.1), fails but Eq. (4.2), holds.

Seller mode is unchanged: M sells only its own product.

Dual Mode: Recommendation Decision

What does M compare?

- Own expected margin from selling M's product or imitation.
- Expected commission from showing S.

Recommendation rule

- M choose to hide S
  - M's expected margin > expected commission from S;
  - Or the marketplace-purchase condition for S, Eq. (4.1), fails.
- M choose to show S
  - M's expected margin < expected commission from S;
  - and the marketplace-purchase condition for S, Eq. (4.1), holds.

Pricing Without Imitation

Without imitation, the subgame can be exploitative or price-squeezing.

1. **Exploitative**: M hides S, and charges the highest price allowed by outside options.
2. **Price squeeze**: M shows S, but caps S's inside price and margin.

Exploitative equilibrium

- **Exploitative price**:

$$p_m^* = \min\{\tau, b\} + \sigma \tag{4.3}$$

  - M vs fringe through marketplace: $v + \sigma + b - p_m^* > v + b - \tau \Longrightarrow p_m^* < \tau + \sigma$
  - M vs fringe not through marketplace: $v + \sigma + b - p_m^* > v \Longrightarrow p_m^* < b + \sigma$

- **Exploitative profit**: using the seller-mode profit function, Eq. (3.1), and Eq. (4.3):

$$\begin{aligned}\Pi_{no-imi}^{exploit} &= p_m^* \times G(v + b + \sigma - p_m^*) \\ &= (\min\{\tau, b\} + \sigma) G(v + b - \min\{\tau, b\}).\end{aligned} \tag{4.4}$$

Pricing Without Imitation: Price Squeeze

Without imitation, products and quality gaps are the same as in the baseline. The only new instrument is steering:

- If S is shown, pricing follows the baseline price-squeeze logic.
- Self-preferencing adds one more constraint: M must prefer showing S to hiding it.

Pricing With Imitation [1]

M can sells the **copied product ($\sigma = \Delta$)** to all consumers at

Exploitative equilibrium

$$\begin{aligned} p_m^* &= \min\{\tau, b\} + \sigma \qquad \text{(imitation: } \sigma = \Delta) \\ &= \min\{\tau, b\} + \Delta. \end{aligned} \tag{4.5}$$

$$\begin{aligned} \tilde{\Pi}_{imi}^{exploit} &= p_m^* \times G(\nu + b + \Delta - p_m^*) \\ &= (\min\{\tau, b\} + \Delta) G(\nu + b - \min\{\tau, b\}). \end{aligned} \tag{4.6}$$

Price-squeeze equilibrium

$$\begin{aligned} p_i^* &= p_m^* + \Delta - \sigma \qquad \text{(imitation: } \sigma = \Delta) \\ &= \min\{\tau, b\} = \tau. \end{aligned} \tag{4.7}$$

notice: if M can imitation $p_i^* = \tau$ does not depend on Δ, S will not innovate.

[1] Imitation matters when S's product is better than M's original product $\Delta > \sigma$.

Proposition 5: Dual Mode Equilibrium

Assumptions in the regulated-practice environment

- **Imitation allow:** M can perfectly imitate S's innovation at no cost.
- **Self-preferencing allow:** M can perfectly steer consumers toward its own offer.
- **Dual mode is feasible:** M both sells its own product and hosts S.

Main variables in equilibrium

- M's profit: $\Pi^{dual} = p_m^* G(\nu + b + \Delta^{dual} - p_m^*) = (b + \max\{\sigma, \Delta^l\})G(\nu)$.
- S's operating profit: $\pi^{dual} = (p_i^* - \tau)G(\nu + b + \Delta^{dual} - p_i^*) = 0$.
- Innovation level: $\Delta^{dual} = \Delta^l \quad \leftarrow \quad \Delta^{dual} = \arg\max_{\Delta^{dual} \geq \Delta^l} \pi^{dual} - K(\Delta)$.
- Consumer surplus: $CS^{dual} = CS^0 \equiv \int_0^\infty \max\{\nu, \nu_o\}\, dG(\nu_o)$.
- Total welfare: $W^{dual} = \Pi^{dual} + \pi^{dual} + CS^{dual}$

Proposition 5: Outcome Vector

Table: Dual-mode benchmark with imitation and self-preferencing

Case	M's choice of mode	Π	π	Δ	CS	W
$\sigma \geq \Delta^l$	Dual - no imitation, M sells	$(b+\sigma)G(\nu)$	0	Δ^l	CS^0	$\Pi + CS^0$
$0 < \sigma < \Delta^l$	Dual - high fee or imitation	$(b+\Delta^l)G(\nu)$	0	Δ^l	CS^0	$\Pi + CS^0$
$\sigma \leq 0$	Dual - high fee or imitation	$(b+\Delta^l)G(\nu)$	0	Δ^l	CS^0	$\Pi + CS^0$

High-fee and imitation paths

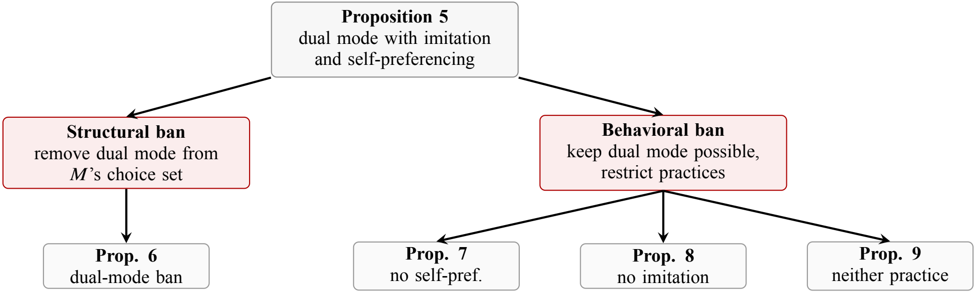
- **High fee** (M as marketplace): $\tau = b + \Delta^l$.
- **Imitation** (M as seller): $p_m = b + \Delta^l$.

- $\pi = 0$: with a high fee, M extracts all of S's price margin; with imitation, M makes all sales itself.
- $\Delta = \Delta^l$: since S's operating profit does not increase with Δ, it has no incentive to innovate above the minimum level.

Policy Map: How the Bans Are Related

Ban types

- **Structural ban**: change M's business model by removing dual mode; it can shift M to a pure seller or pure marketplace mode.
- **Behavioral ban**: keep dual mode possible but remove specific practices; it can restore price discipline and/or innovation incentives.



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Dual-Mode Ban – Step 1: What Is Banned?

Policy object

The ban removes the platform's ability to operate in dual mode within this product category.

- **Economic meaning**: M cannot both host S on its marketplace and sell its own competing product in the same category.
- **What it does not directly ban**: product imitation or self-preferencing as practices.
- **Mathematical meaning**: remove dual mode from M's stage-0 choice set,

$$\{\text{Marketplace}, \text{Seller}, \text{Dual}\} \quad \longrightarrow \quad \{\text{Marketplace}, \text{Seller}\}.$$

Post-ban mode choice

$$M \text{ chooses } \arg\max\{\Pi^{mkt}, \Pi^{sell}\}. \tag{4.8}$$

Dual-Mode Ban – Computing the Post-Ban Outcome

Once dual mode is removed, M compares the two pure modes.

Mode	Π	π	Δ	CS	W
Seller	$(b+\sigma)G(\nu)$	0	Δ^{l}	CS^{0}	$(b+\sigma)G(\nu)+CS^{0}$
Marketplace	$bG(\nu)$	$\Delta^{mkt}G(\nu)-K(\Delta^{mkt})$	Δ^{mkt}	CS^{0}	$(b+\Delta^{mkt})G(\nu)-K(\Delta^{mkt})+CS^{0}$

Mode selected after the ban

If $\sigma>0$, M chooses seller mode; if $\sigma\le 0$, M chooses marketplace mode.

Dual-Mode Ban – Step 2: Equilibrium Result

Proposition 6. Compare the post-ban pure-mode outcome with the Proposition 5 dual-mode benchmark.

Region	M's choice of mode	Π	π	CS	Δ	W
$\sigma \geq \Delta^l$	Seller	=	=	=	=	=
$0 < \sigma < \Delta^l$	Seller	↓	=	=	=	↓
$\sigma \leq 0$	Marketplace	↓	↑	=	↑	↑

Interpretation

A structural ban changes M's business model, but it does not directly restore on-platform competition or innovation incentives.

Alternative Policy: Behavioral Remedies

Instead of banning the dual role itself, the regulator can target specific conduct within dual mode.

Remedy	What is banned	Main channel restored
Ban self-preferencing only	M must show S when S is listed	on-platform competition; showrooming constraint
Ban imitation only	M cannot copy S's innovation	S's innovation incentive, if commitment is valuable
Ban both practices	no steering and no copying	baseline dual-mode mechanisms

Logic

For each behavioral remedy, use the same two-step structure: identify the banned conduct, then compare the resulting equilibrium with Proposition 5.

Self-Preferencing Ban – Step 1: What Is Banned?

Policy object

M must always disclose/show S's product whenever S participates on the marketplace.

- **Economic meaning**: M cannot steer consumers away from S; consumers observe all on-platform offers.
- **What remains allowed**: product imitation by M.
- **Mathematical meaning**: the showrooming constraint is restored, so M cannot set a fee above the outside-channel discipline:

$$\tau \le b.$$

Post-intervention dual-mode outcome

$$\tau = b, \qquad \Delta = \Delta^l, \qquad \Pi = (b + \max\{\sigma - \Delta^l, 0\})G(\nu + \Delta^l), \qquad \pi = 0. \tag{4.9}$$

Self-Preferencing Ban – Step 2: Equilibrium Result

Proposition 7. Let $\underline{\sigma}^{steer}$ be the cutoff at which M is indifferent between seller mode and dual mode without self-preferencing.

Region	M's choice of mode	Π	π	CS	Δ	W
$\sigma \geq \Delta^l$	Seller	$=$	$=$	$=$	$=$	$=$
$\underline{\sigma}^{steer} < \sigma < \Delta^l$	Seller	$\downarrow$	$=$	$=$	$=$	$\downarrow$
$\sigma \leq \underline{\sigma}^{steer}$	Dual, no self-pref.	$\downarrow$	$=$	$\uparrow$	$=$	$\uparrow$

Interpretation

This ban restores price competition, but not innovation: imitation still keeps S's equilibrium profit at zero.

Imitation Ban – Step 1: What Is Banned?

Policy object

M is prohibited from copying S's product innovation.

- **Economic meaning**: M can credibly commit not to imitate, so S's innovation may become valuable again.
- **What remains allowed**: self-preferencing/steering by M.
- **Mathematical meaning**: M can choose a fee that respects the innovation constraint

$$\tau \leq \bar{\tau},$$

where $\bar{\tau}$ is the fee cutoff that keeps S willing to choose high innovation.

Imitation Ban – Step 2: Equilibrium Result

Proposition 8. The ban has an effect only when it induces M to set $\tau = \bar{\tau}$ and preserve S's innovation incentive.

Condition for the innovation-preserving equilibrium

$$b + \Delta^l \leq \bar{\tau} \quad \text{or} \quad \bar{\tau} G(\nu + \sigma + b) \geq (b + \max\{\sigma, \Delta^l\}) G(\nu). \tag{4.10}$$

Region	M's mode	Π	π	CS	Δ	W
$b + \Delta^l > \bar{\tau}$ and condition fails	Dual, no imitation	=	=	=	=	=
$b + \Delta^l \leq \bar{\tau}$ or condition holds	Dual, no imitation	$\uparrow$	=	$\uparrow$	$\uparrow$	$\uparrow$

Ban Both Practices – Step 1: What Is Banned?

Policy object

M is prohibited from both self-preferencing and product imitation.

- **Economic meaning**: consumers see S's product, and S is protected from copying by M.
- **Mathematical meaning**: the model returns to the baseline dual-mode mechanism from Proposition 3.
- The showrooming constraint disciplines τ, and the absence of imitation can restore S's innovation incentive.

Ban Both Practices – Step 2: Equilibrium Result

Proposition 9. Let $\underline{\sigma}$ be the baseline cutoff at which M switches between seller mode and dual mode after both practices are banned.

Region	M's choice of mode	Π	π	CS	Δ	W
$\sigma \geq \max\{\underline{\sigma}, \Delta^l\}$	Seller	$=$	$=$	$=$	$=$	$=$
$\underline{\sigma} < \sigma < \Delta^l$	Seller	$\downarrow$	$=$	$=$	$=$	$\downarrow$
$\sigma \leq \underline{\sigma}$	Dual, no imitation or self-pref.	$\downarrow$	$\uparrow^\dagger$ or $=$	$\uparrow$	$\uparrow^\ddagger$ or $=$	$\uparrow$

- $\dagger$: π increases if $b < \bar{\tau}$; otherwise it is unchanged.
- $\ddagger$: Δ increases if $b < \bar{\tau}$ or condition (8) holds; otherwise it is unchanged.

Interpretation

Banning both practices restores both channels: price discipline from visibility and innovation incentives from no imitation.

Policy Comparison

- **Structural ban**: removes dual mode, but does not directly fix imitation or self-preferencing.
- **Ban self-preferencing**: restores price competition, but innovation remains low because imitation is still possible.
- **Ban imitation**: can restore innovation when M benefits from committing not to copy.
- **Ban both**: closest to the baseline dual-mode mechanism and can restore both price competition and innovation.

Policy lesson

The harmful conduct is often a better regulatory target than the platform's dual role itself.

Section 4 Takeaways

1. Perfect imitation and perfect steering make dual mode weakly best for M.
2. In dual mode, S chooses Δ^l because innovation rents are extracted.
3. A dual-mode ban does not directly fix copying or steering.
4. Banning self-preferencing restores price competition but not innovation.
5. Banning imitation restores innovation when commitment is valuable enough.
6. Banning both practices restores the baseline dual-mode mechanisms.

Main policy message

The harmful conduct is often a better regulatory target than the platform's dual role itself.

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PART 5

Policy and Extensions

Presenter: Zhuoyue Wu

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